

15.2 Double Integrals over General Regions

1 Warm Up

Problem 1: Graph the following functions. $y = (x - 1)^2$, $y = \sqrt{x}$, $y = \sin x$.

Problem 2: Find $\int \ln x$, $\int \cos^2 x$, $\int x\sqrt{1 - x^2}$.

2 Motivation

In the previous section we defined a double integral which is the volume under a surface bounded by a rectangle. In words, the way we define this is to cut up our rectangle into infinite smaller rectangles and by selecting an arbitrary point inside the rectangle we can find a height for each rectangle and add up the volumes of infinitely many rectangular prisms. What happens though when we want the volume under a bounded region which is not a rectangle. In this section we answer this!

3 Definitions & Theorems

1. In order to find a double integral or the volume under a surface we need a surface $f(x, y)$ and a region R . It is important not to confuse what is a region and what is the surface.
2. If R is a bounded region which is not a rectangle, we can imagine our region being enclosed by a rectangle. From last section we know that we can chop up our rectangle into infinitely many smaller rectangles and find the volume of the rectangular prisms. We have one caveat. Only choose the rectangles in with the arbitrary point lies within the original region R . Thus, when the number of rectangles approaches infinity, the rectangular prisms which we choose will get finer and finer towards the bounded region R .
3. The general form of our double integrals look like this when our region is bounded by $a \leq x \leq b$ and $g_1(x) \leq y \leq g_2(x)$.

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

4. The general form of our double integrals look like this when our region is bounded by $h_1(y) \leq x \leq h_2(y)$ and $c \leq y \leq d$.

$$\int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

5. Here are a few helpful hints in order to know the order of integration.
- The bounds of the outer integral must be constants in order for our final answer to be a number (volume is a number)
 - If I am integrating with respect to dx I am moving across the x -axis from left to right and my bounds should be $x = f(y)$ bounds.
 - If I am integrating with respect to dy I am moving across the y -axis from bottom to top and my bounds should be $y = f(x)$.
 - I *never* want to switch functions as I move across my axis between my bounds.
 - Look at the integral and try to make your life easier (look for u-sub)
6. We will now see a TON of example regions and integrals to demonstrate the various set-ups we may encounter and how to choose which order of integration I should choose.

4 Examples

Problem 1. Evaluate $\iint_R 2xy \, dA$ where $R = \{(x, y) : 0 \leq x \leq 1/2, 0 \leq y \leq \sqrt{1-x}\}$

Problem 2. Set up $\iint_R f(x, y) \, dA$ where R is bounded by $y = 2x$ and $y = x^2$.

Problem 3. Evaluate $\iint_R \frac{x}{1+xy} \, dA$ where $R = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}$

Problem 4. Evaluate $\iint_R \frac{\ln y}{y} \, dA$ where $R = \{(x, y) : 0 \leq x \leq \pi, e^{-2x} \leq y \leq e^{-\cos x}\}$

Problem 5. Evaluate $\iint_R ye^{xy} \, dA$ where $R = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\}$

Problem 6. Evaluate $\iint_R \frac{1}{xy} \, dA$ where $R = \{(x, y) : 1 \leq y \leq e, y \leq x \leq y^2\}$

Problem 7. Evaluate $\iint_R (\sin x - y) \, dA$ where R is bounded by $y = \cos x$, $y = 0$, $x = 0$, and $x = \pi/2$

Problem 8. Evaluate $\iint_R 4x^3 \, dA$ where R is bounded by $y = (x - 1)^2$ and $y = -x + 3$

Problem 9. Suppose we are given the following integral which cannot be solved using elementary integration methods. Switch the order of integration and evaluate.

$$\int_0^1 \int_2^{y^2} e^{-x^2} \, dx \, dy$$

Problem 10. Suppose we are given the following integral which cannot be solved using elementary integration methods. Switch the order of integration and evaluate.

$$\int_0^4 \int_{\sqrt{x}}^2 \sin y^3 \, dy \, dx$$

5 Scratch Work